

Each pixel on the screen is made up of a combination of three colors: red, green, and blue, often referred to as *RGB*. The color displayed in any pixel is composed of some amount of red, green, and blue, where the amount of each is specified as a value from 0, meaning none, to 255, meaning full intensity. Therefore, there are $256 \times 256 \times 256$ possible combinations, or 16,777,216 (often referred to as just “16 million”) possible colors, for each pixel.

Colors in pygame are given as RGB values, and we write them as Python tuples of three numbers. Here is how we create constants for the main colors:

```
RED = (255, 0, 0)    # full red, no green, no blue
GREEN = (0, 255, 0)  # no red, full green, no blue
BLUE = (0, 0, 255)   # no red, no green, full blue
```

Here are the definitions of a few more colors. You can create a color using any combination of three numbers between 0 and 255:

```
BLACK = (0, 0, 0)    # no red, no green, no blue
WHITE = (255, 255, 255) # full red, full green, full blue
DARK_GRAY = (75, 75, 75)
MEDIUM_GRAY = (128, 128, 128)
LIGHT_GRAY = (175, 175, 175)
TEAL = (0, 128, 128) # no red, half-strength green, half-strength blue
YELLOW = (255, 255, 0)
PURPLE = (128, 0, 128)
```

In pygame, you’ll need to specify colors when you want to fill the background of a window, draw a shape in a color, draw text in a color, and so on. Defining colors up front as tuple constants makes them very easy to spot later in code.

Event-Driven Programs

In most of the programs in the book so far, the main code has lived in a `while` loop. The program stops at a call to the built-in `input()` function and waits for some user input to work on. Program output is typically handled using calls to `print()`.